

Axon File Transfer

What is it?

Axon is a desktop tool for transferring data (files/text/URLs) across a local network, particularly between PCs and mobile devices. Axon's goal within this niche is to make the basic transfer process as quick and effortless as possible. It also strives to be highly intuitive.

Outbox

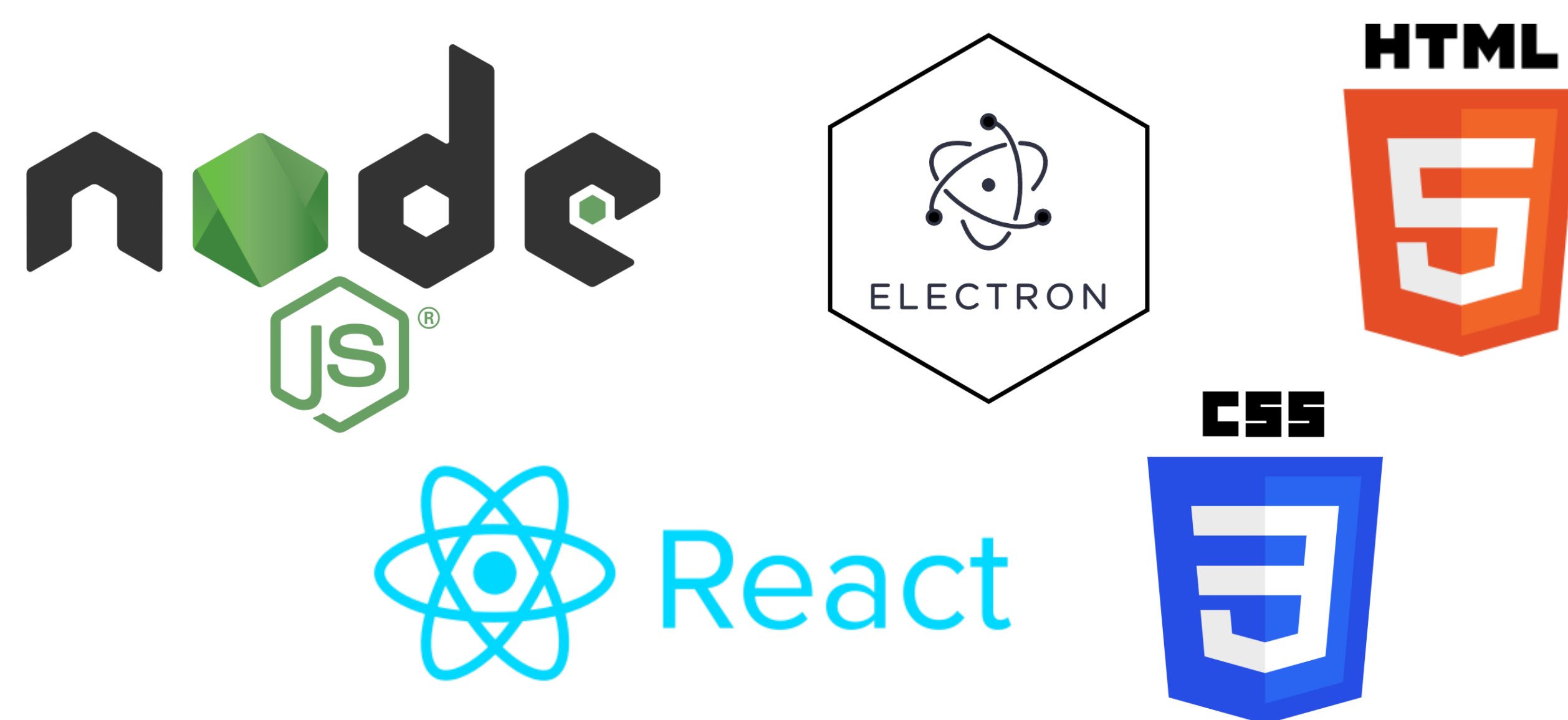
The Outbox allows for items to be sent from Axon to other devices. Items can be added in a variety of ways. Files can be drag-and-dropped into the Axon GUI, or a file selector can be opened to locate files manually. Text can also be added via an input field.

Inbox

The Inbox allows for Axon to receive items from other smart devices. For example, a smartphone just needs to scan the Inbox QR code or enter the URL and then can send items through the intuitive web interface.

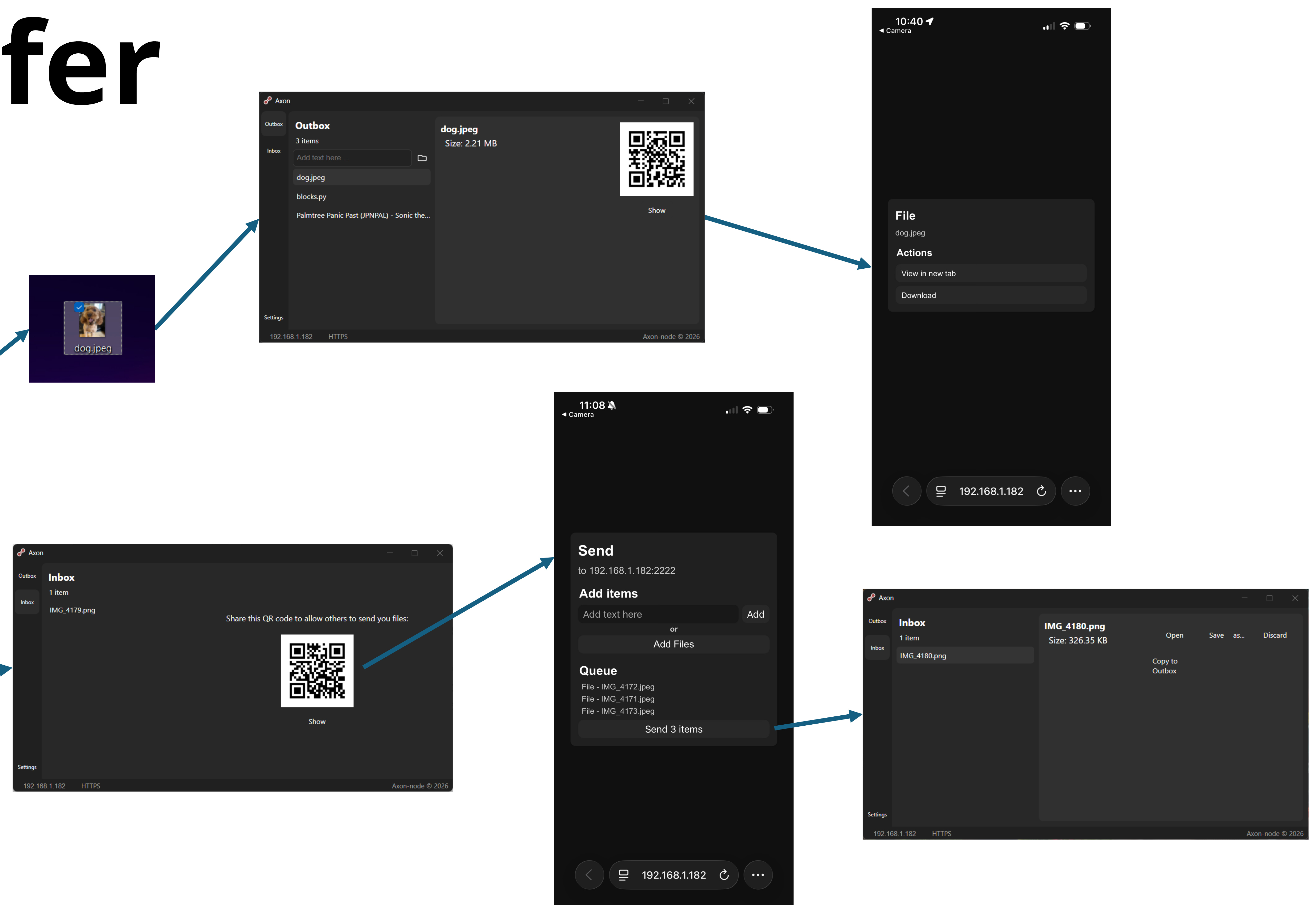
Tech Stack

Axon is built entirely in JavaScript with the ElectronJS framework, which includes a NodeJS runtime and an instance of the chromium browser to manage the user interface. The React library was also used to streamline the HTML/CSS building process.



Development

I am the sole developer of Axon. I do everything from designing the server behavior to the UI/UX. Progress is tracked and organized on the project's GitHub repository. Following the first beta release earlier this year, I received helpful feedback from friends and family about bugs and missing features, which helped to guide my work this semester.



Results

This semester, I finished implementing the core functionality of the Inbox and Outbox for both files and text. I also redesigned the frontend-backend communication structure to make it more maintainable long-term. I also spent a significant amount of time on the UI, making each section as visually consistent as possible, and implementing “small” features that make the experience feel polished and intuitive. These included:

- Settling on a color palette and a set of style rules that are now used throughout all user interfaces.
- Overhauling the client Inbox web interface with AJAX, and introducing a queue system so users can easily send items in batches
- Adding relevant Inbox actions for after an item has been received (Open, Reveal, Save As, Copy to Outbox)
- Overhauling the Settings menu with improved data validation and a new visually consistent and intuitive design.
- Various visual bugs

Conclusions

This semester was spent mostly improving, refactoring, and tweaking an existing codebase, in contrast to last semester which involved more rapid development of new features. This has taught me a lot about how I work, and in turn how I can use my time more efficiently. In line with that, I spent much more time this semester focusing on minor details in pursuit of a highly polished user experience, which I did not concern myself with last semester. I realized this can turn into a major time sink, and I've become much more effective at knowing when something is “good enough” and isn't worth spending time on anymore. I have really enjoyed working on this project and I look forward to continuing to make it as useful as possible.



GitHub Repo